

Final Test Report:

Homburg Appointment Booking System

1. Executive Summary

This report summarizes the validation results for the Homburg appointment booking project. The testing suite provides comprehensive coverage across functional, UI/UX, accessibility (BITV/WCAG), and negative scenario domains.

Total Test Coverage:

- **Automated Scenarios:** 19
- **Manual Test Cases:** 4
- **Audit Checklists:** 22 specific validation points (UI/UX & Accessibility)

Overall Status: Stable (Production Ready with Known Issues).

Execution Environment: Chrome, Firefox, Safari (Desktop & Mobile).

2. Test Scope & Methodology

A. Automated Regression (Playwright)

The automated suite covers critical paths and system edge cases:

- **Core Flows:** Successful booking scenarios for "Bürgeramt" and "Fahrerlaubnis".
- **Logic & UI:** Service counter limits, filter interactivity, dynamic overview synchronization, and header/footer consistency.
- **Negative/Security:** Deep linking protection, session management (timeout/warning), form validation (input security).

B. Manual Auditing (Checklists)

The system was validated against high-standard quality requirements:

- **UI/UX & Responsive:** Layout integrity on 430px viewports, touch-target sizing, and modal behavior.
- **Accessibility (BITV 2.0 / WCAG 2.1):** Full keyboard navigation (Tab order, focus traps), screen reader semantics (ARIA labels, dynamic states), and contrast ratios.
- **Integration:** Validated external links and session isolation during modal interactions.

3. Execution Results

Test Category	Total Items	Passed	Failed/Blocked
Automated Scenarios	19	17	2 (1 Blocker, 1 Major Defects)
Manual Checklist Blocks	4	3	1 (1 Minor / Major)
Audit Checklists	22	22	0
TOTAL	45	42	3

4. Discovered Critical Issues (Bug Reports)

During testing, two major validation defects were identified that impact data integrity.

ID	Title	Severity	Impact
BUG_01	Invalid special characters accepted in 'Jahr' field	Minor / Major	Allows invalid data entry, potentially causing database corruption.
BUG_02	Inconsistent email validation across input fields	Critical	Prevents appointment confirmation delivery; renders service unusable.
BUG_03	UI displays a blank page when Calendar API returns 500 error	Critical	Disrupts the booking flow; user unable to select a date, leads to bounce.

5. Accessibility & Responsive Compliance

The project underwent a rigorous audit following BITV 2.0 / WCAG 2.1 standards:

- **Responsive Integrity:** Confirmed 0 horizontal scrolling at 430px (Mobile).
- **Keyboard Accessibility:** Confirmed full navigability (Tab Order, Focus Traps in Modals, Escape Key functionality).
- **Semantics:** Verified ARIA states for key dynamic components (accordions, service counter buttons) to ensure proper screen reader feedback.

6. Conclusion & Recommendations

The system demonstrates high functional stability, with comprehensive test coverage across critical user flows, responsive design, and accessibility standards (BITV 2.0 / WCAG 2.1). Manual audits confirm full compliance with accessibility requirements, ensuring an inclusive user experience.

However, to achieve **production-grade reliability**, the following areas require remediation:

- **Data Integrity & Validation:** The critical validation defects (BUG_01, BUG_02) identified in the personal data step (Step 5) must be resolved to prevent data corruption and ensure the successful delivery of appointment confirmations.
- **System Resilience & UX:** A critical UX defect (BUG_03) was identified regarding API error handling. The application currently fails to display user-friendly notifications when the Calendar API returns an error, resulting in a blank UI state. Implementing clear error messaging is essential to improve system transparency and meet modern UX/UI standards.

Addressing these issues is vital to ensure service reliability and a seamless booking experience for all end-users.